Sleep Cycle

Different wave frequencies:

1) Beta 13-30 Hz Awake, alert, thinking actively
2) Alpha 8-13 Hz Relaxed, eyes closed, calm but awake
3) Theta 4-8 Hz Light sleep, meditation, early sleep stages
4) Delta 0.5-4 Hz Deep sleep (N3) = the slowest, strongest waves

The cycle:

1) Awake
2) N1 - Light Sleep
3) N2 - Light Sleep still but deeper
4) N3 - Deep Sleep (Delta waves)
5) N2 - (again)
6) REM - Dreaming, brain active

Cycle repeats 4-6 times per night
Early night = more deep sleep (N3)
Late night = more REM sleep
A full night: 20-25% REM, 20-25% deep sleep, rest is N1 + N2

Note: REM sleep does not correspond to a single brain-wave frequency; instead, it is characterized by low-amplitude, mixed-frequency activity(mainly theta, alpha, and beta) making the brain appear awake while the body remains asleep. Occurs usually later at night?

Importance of REM:

- External input is reduced so brain can replay emotional and memory-related activity
- Synapses are selectively strengthened or weakened supporting emotional memory processing
    - Long-Term Potentiation(LTP) = Strengthening → good for learning/mem.
    - Long-Term Depression(LTD) = Weakening → “fades out” the unnecessary
- REM deprivation impairs synaptic plasticity in areas like hippocampus/prefrontal cortex
    - Synaptic plasticity is the brain’s ability to strengthen or weaken connections between neurons, allowing learning, memory formation, emotional regulation, and adaptation
- REM rebound=if REM is disrupted then response is entering it faster/longer/ more often
    - Short term this is good
    - Long term bad: often comes with fragmented sleep & reduce overall sleep quality
    - For the project this means we should emphasize our attempts to reduce REM disruption and why that is so necessary
- REM measured by latency, duration, density (eye movement frequency), and continuity
- Fragmented REM disrupts: Emotional regulation/Fear extinction/Memory consolidation
    - This is why *stability is more important than raw quantity*
- Problems causing poor REM include:
    - Stress
    - Circadian misalignment

    - Sleep restriction
    - Environmental disturbance (noise, light, temperature)
    - Irregular sleep timing

Neuroplacicity = process that includes “adaptive structural and functional changes to the brain” → this basically means it affects changes in things like memory, learning, mood, and sleep.

Electroencephalogram(EEG) = sensor for scalp used to measure brain activity (this includes synchronized activity of millions of neurons, specifically, electrical signals from pyramidal neurons in the cortex.) → What does this mean? We can measure frequency/amplitude of brain waves to identify sleep stages(alpha, beta, ect…)

- REM is identified using EEG when:
    - EEG shifts from synchronized (NREM) → desynchronized
    - Delta activity drops sharply
    - Theta remains present but becomes irregular
    - Low amplitude measured
- However EEG is not really enough to know its REM. In addition to EEG we should also use HRV

Heart Rate Variability(HRV) = exactly what is sounds like → HRV is healthy

****PPG(optical heart sensors)** - can measure HRV

Before asleep	NREM	REM	Awakenings
lower/irregular HRV	High/stable HRV	HRV shows bursts (but follows a pattern)	Sudden HRV drop

Muscle Atonia = muscles basically paralyzed (happens during REM) → this cannot be measured directly using a wearable device, however it can be inferred with sensors above

<u>Impact of light:</u>

- Light influences sleep indirectly (through circadian phase shifts) and directly (modulating sleep onset and homeostatic pressure).
- Disruption of light patterns (shift work, jet lag, constant light) can desynchronize circadian rhythms and impair sleep architecture (reduced slow-wave sleep, altered REM latency).
- Because light affects both systems (circadian/non-circadian), getting the right amount, colour, and timing of illumination is essential for:
    - Mental health (preventing or easing depression, anxiety)
    - Cognitive performance (attention, memory, learning)
    - Overall well-being (energy, sleep quality, stress levels)

Impact of sound:

- N3, also called slow-wave sleep (SWS), is the deepest stage of non-REM sleep and is defined by dominant delta-frequency brain activity.
- quiet sounds delivered in phase with the brain's own slow oscillations during deep NREM sleep can strengthen those oscillations and improve memory consolidation (but must be timed right)
    - Not helpful for our headband because it will hurt more then help if not timed exactly
- Pink noise has the best effects, if can help with:
    - more slow-save activity (SWA) during NREM sleep - amount of low-frequency, high-amplitude brain activity (0.5 - 4 Hz) seen on EEG during N3 / SWS(slow wave activity)
    - Larger sleep spindle amplitude - the voltage of brief bursts of brain activity in the 12 - 15 Hz range that occur mainly during NREM sleep
    - Decreased sleep onset latency = falling asleep faster
    - Less micro-arousals, especially during REM = greater sleep stability
    - In the study ppl said the felt deeper sleep, fewer awakenings, faster relaxation before sleep

Common dimensions:

Units: mm

Fit / sizing

- HeadCirc_Min = 520
- HeadCirc_Max = **620**
- Overlap_Min = **30** *(minimum strap overlap so it can't slip out)*
- Overlap_Max = **120** *(max overlap when tightened)*

Band geometry

- BandWidth = **38**
- BandThickness = **2.5**
- PaddingThickness = **4.0** *(soft liner, optional body)*
- EdgeFillet = **2.0**

Arc shape

- BandRadius = **90** *(forehead curvature radius)*
- BandArcAngle = **210 deg** *(wrap around forehead/sides)*

Strap / adjust

- StrapWidth = **30**
- StrapThickness = **1.6**
- StrapTravel = **100** *(how much circumference adjust you get via slider)*

Forehead sensor pod

- PodW = **34**
- PodH = **26**
- PodD = **10**
- PodWall = **1.6**
- PodFillet = **3**

- SensorPadW = **22**
- SensorPadH = **18**
- SensorPadProtrude = **2.0** *(contact bump)*

Rear battery pod

- RearPodW = **45**
- RearPodH = **28**
- RearPodD = **12**
- RearPodWall = **1.6**
- RearPodFillet = **4**

Vibe (massage) motor pockets (optional)

- MotorDia = **10.0**
- MotorDepth = **3.5**
- MotorCountEachSide = **2**
- MotorSpacing = **30**